ASSIGNMENT BRIEF

HTU Course No: 10203340	HTU Course Name: Cryptography
BTEC Unit Code:	**BTEC UNIT Name:**

Student Name/ID Number/Section	
HTU Course Number and Title	10203340 Cryptography
BTEC Unit Code and Title	
Academic Year	2024-2025 3
Assignment Author	Elham Derbas
Course Tutor	Elham Derbas
Assignment Title	Lock&Key
Assignment Ref No	1
Issue Date	05/08/2025
Formative Assessment dates	From 06/08/2025 to 28/08/2025
Submission Date	14/09/2025
IV Name & Date	Sami Almashaqbeh 04/08/2025

Submission Format

Student is expected to individually submit his/her work including:

- An individual written report in **Word format** covering the required details in the (Assignment Brief and Guidance) section.
- A signed Student Assessment Submission and Declaration form (A test conducted on the e-learning system can fulfill this requirement).
- Evidence of the implemented work (softcopy). Students should submit their original source code.
- An oral discussion about the report and the implemented work. Instructions, date, and time for the discussion will be provided later. A witness statement or observation record is considered as evidence for this part.
- An in-class written exam.

PS: Files should be uploaded separately rather than in a zipped file.

Report guidelines:
The report should be written in a concise, formal business style using single spacing and font size 12 with use of headings, paragraphs and subsections as appropriate (Cover page, table of contents, and an introduction to provide an overview of your report.). The expected word limit is about 3000 words, although you will not be penalised for exceeding the total word limit. The report must be supported with research and referenced using the Harvard referencing system.

Note: Soft copies submissions should be done through the university's eLearning system (https://elearning.htu.edu.jo) by the deadline assigned above.

Unit Learning Outcomes

LO1 Apply mathematical concepts in cryptographic applications.

LO2 Examine the symmetric encryption ciphers and algorithms.

LO3 Assess public key encryption protocols and digital signatures.

LO4 Assess the requirements for secure message exchange.

Assignment Brief and Guidance

You have been appointed as a cryptography specialist for a project involving the development of a tool named "Lock & Key" a secure communication system designed to ensure confidentiality, integrity, and authentication. Your task is to design and implement the system according to the specified business requirements. To achieve this, you are expected to complete a series of assigned tasks that collectively contribute to building a fully functional solution. You must submit a comprehensive report detailing your approach, the steps taken for each task, and all required information as outlined.

Your system should display the following main menu, allowing the user to select the desired service:

***** LOCK && KEY *****

1. Confidentiality Only: in this mode, the message payload must be encrypted using a symmetric cipher. You are required to implement the DES (Data Encryption Standard) algorithm from scratch to support this functionality.

2. Authentication Only:This mode focuses on digital signature generation to ensure the authenticity of the transmitted vote. Your system must allow verification of the sender's identity.

3. Confidentiality & Authentication: This mode combines both encryption and digital signatures to ensure that messages are both confidential and authenticated

4. RSA cryptanalysis.

5. Image encryption.

Your Lock and Key implementation must support all operational modes. While you have the flexibility to choose the hash functions used in your design, RSA is mandatory for all asymmetric cryptographic operations. Additionally, your system must include a secure mechanism for key generation and exchange to ensure both the sender and receiver can communicate safely.

Task 1:

You have been assigned by your manager to develop your own code—using any programming language of your choice—to implement the confidentiality utility of the Lock and Key tool. This utility must enable a text message to be encrypted and decrypted. The sender is responsible for manually providing the shared secret key required for the symmetric encryption process (In hexadecimal format). To enhance security, your implementation must use Cipher Block Chaining (CBC) mode and implement ciphertext stealing to correctly handle message lengths that are not a multiple of the encryption algorithms block size.

To demonstrate that your code functions correctly, a test message along with its expected ciphertext will be provided via the e-learning platform. You will be required to present your fully functional tool during the oral examination.

Deliverable:

- Your code (in text format).

- Include screenshots of the results obtained from running your code, and document them in a report. The report should contain clear descriptions of the test cases, including the input, expected output, and actual output, to demonstrate and validate the correctness of your implementation.
- An explanation of how the CBC mode was applied in your work.

Task 2:

To ensure the second utility functions correctly, your program should first inform the user that RSA key generation is required before proceeding. It must then prompt the user to input two prime numbers (10 digits each) and a value for the public exponent (e) to facilitate RSA key generation.

Once the keys are generated, the program should prompt the user to enter a message to digitally sign. Your implementation must also include functionality to verify the digital signature, ensuring the integrity and authenticity of the message.

For hashing, you may use any built-in hash function, provided it meets standard cryptographic hash function criteria (e.g., pre-image resistance, collision resistance, and second pre-image resistance).

Deliverable:

- Your code (in text format).
- Include screenshots of the code execution results along with clear, step-by-step documentation of your implementation process.
- The parameters used for generating the asymmetric keys.

NOTE: The third utility in our tool simply combines the functionality of the first two utilities in one place for easier use.

Task 3:

Your team has been tasked with evaluating the suitability of using the RSA encryption system not only for key generation and digital signatures but also for directly encrypting short messages. You are specifically asked to analyze and justify whether RSA can be securely used for short message encryption. As part of this task, you are expected to explore potential vulnerabilities, such as short message attacks on RSA, and assess whether they pose a practical threat in this context. To support your justification, you must implement a code-based demonstration showing RSA encryption of short messages and evaluate the results. This will help prove or disprove the practicality and security of using RSA for encrypting short, plaintext messages in your system. For testing purposes, you may use any two-digit values for n.

Deliverable:

- The detailed step-by-step you followed in your test.
- Provide a detailed explanation of the security considerations when using RSA for encrypting short M essages.
- Include a screenshot of the output from your security assessment, demonstrating the results of your RSA short message attack test and any related vulnerability checks.

- Your code/scripts (in text format).

Task 4:

The last utility is about providing users with the ability to encrypt and decrypt images, to implement that you have been tasked with developing a simple image encryption tool.

You proposed to use either ECB (Electronic Codebook) or CBC (Cipher Block Chaining) mode for this purpose. And implement them from scratch. Given your expertise in cryptography, you are required to conduct a test case where you encrypt the provided image file, test_image.jpg, first using ECB mode and then using CBC mode. Both encryptions must apply Ciphertext Stealing (CTS) to handle padding. After completing the test, you are expected to analyze the output images and provide a justification for which encryption mode is more suitable for secure image encryption in the context of the Lock & key tool.

Deliverable:

- The detailed step-by-step you followed in your test.
- An explanation of why you found a specific mode more suitable.
- Include both output images with proper captions.
- Your code/scripts (in text format)

Task 5:

Your manager has asked you to explain the operation of the Square and Multiply algorithm and explain how it was used in your code. Also, he asked for a critical evaluation of your Lock & Key tool and suggestions for future enhancements.

In-class Exam: The exam is scheduled for Aug 15, 2025, at 1:00 PM and will cover the following criteria: P1, P2, P3, P5, P6, M1, and M3.

1) GCDs and the Extended Euclidean algorithm.

2) Birthday problems.

3) Public key cryptography.

4) Key exchange.

5) Multiplicative inverse.

6) Attacks on asymmetric ciphers.

Learning Outcomes and Assessment Criteria			
Learning Outcome	**Pass**	**Merit**	**Distinction**
LO1 Apply mathematical concepts in cryptographic applications.	**P1** Calculate the greatest common divisor of a pair of numbers. **P2** Apply the Extended Euclidean algorithm to find the multiplicative inverse. **P3** Compute probabilities and/or number of hashes needed for certain hash function criteria.	**M1** Apply Euler and Fermat theorems to compute the multiplicative inverse.	**D1** Produce a detailed written explanation of the importance of the Square and Multiply algorithm and its application in asymmetric cryptography
LO2 Examine the symmetric encryption ciphers and algorithms.	**P4** Explain the use of the building blocks in symmetric ciphers.	**M2** Compare between encryption modes.	**D2** Evaluate the effect of error propagation for various encryption modes.
LO3 Assess public key encryption protocols and digital signatures.	**P5** Examine common public key cryptographic methods and their uses. **P6** Explain public key exchange and digital signatures.	**M3** Discuss the common attacks on an asymmetric encryption scheme.	**D3** Critically evaluate the message exchange system and provide suggestion for future enhancements.
LO4 Assess the requirements for secure message exchange.	**P7** Produce code that implements mathematical ciphers and algorithms to encrypt and decrypt data.	**M4** Build a secure message exchange tool that guarantees privacy, integrity, and authentication.	